Pivotal Quantities

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Introduction

A pivotal quantity – or pivot – is a function of the data and the parameter whose probability distribution does not depend on the parameter. The existence of a pivot is what makes exact confidence intervals possible: if $Q(X, \theta)$ has a known distribution, we can rearrange the probability statement about Q into a probability statement about θ, and that rearrangement is the CI.

Prerequisites

The reader should understand the concepts of confidence intervals, the t-distribution, and the chi-squared distribution.

Theory

A **pivotal quantity** $Q = Q(X_1, \ldots, X_n; \theta)$ satisfies: for every θ, the distribution of Q is the same known distribution. This is stronger than "a test statistic under the null": a pivot's distribution does not depend on θ *for every* value of θ.

Canonical examples (iid data):

- **Normal, known** σ^2: $Z = \sqrt{n}(\bar{X} - \mu)/\sigma \sim \mathcal{N}(0, 1)$.
- **Normal, unknown** σ^2: $T = \sqrt{n}(\bar{X} - \mu)/s \sim t_{n-1}$.
- **Normal variance**: $W = (n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$.
- **Ratio of normal variances**: $F = s_1^2/\sigma_1^2 \cdot \sigma_2^2/s_2^2 \sim F_{n_1-1, n_2-1}$.
- **Uniform**$(0, \theta)$: $\max X_i/\theta \sim \mathrm{Beta}(n, 1)$.

Given a pivot Q and its known distribution, pick quantiles $q_{\alpha/2}$ and $q_{1-\alpha/2}$ such that $P(q_{\alpha/2} \le Q \le q_{1-\alpha/2}) = 1 - \alpha$. Invert the inequalities to isolate θ; the resulting set is a $1 - \alpha$ CI with exact coverage.

When no exact pivot exists, **asymptotic pivots** (e.g., $(\hat{\theta} - \theta)/\widehat{\mathrm{SE}}$ which is asymptotically $\mathcal{N}(0, 1)$) give CIs with approximate coverage.

Assumptions

Exact pivots depend on distributional assumptions (normality, uniformity). When those fail, the nominal coverage is not guaranteed; bootstrap methods or robust asymptotic pivots are alternatives.

R Implementation

```r
set.seed(2026)

n <- 25
mu <- 5
sigma <- 2

x <- rnorm(n, mu, sigma)
```

```r
xbar <- mean(x)
s    <- sd(x)
t_crit <- qt(0.975, df = n - 1)

ci_mean <- xbar + c(-1, 1) * t_crit * s / sqrt(n)
ci_mean

chi_l <- qchisq(0.025, df = n - 1)
chi_u <- qchisq(0.975, df = n - 1)
ci_var <- c((n - 1) * s^2 / chi_u, (n - 1) * s^2 / chi_l)
ci_var
```

Two exact CIs are constructed: the t-pivot for μ, and the chi-squared pivot for σ^2.

Output & Results

```
[1] 4.272  6.044     # 95% CI for mu
[1] 2.197  7.115     # 95% CI for sigma^2 (true value 4)
```

Both intervals are built from pivots with exact distributions under the normal model; their coverage is exactly 95% across repeated sampling.

Interpretation

Pivotal quantities are often implicit in textbook CIs. Reporting a t-interval for a mean or a chi-squared interval for a variance is reporting a pivot-based construction. The pivot formalism explains why these CIs have exact coverage under the assumed model – and why they lose coverage when the model is wrong.

Practical Tips

- When an exact pivot exists, prefer it over an asymptotic normal approximation, especially for small n.
- Pivots are rare outside the normal family; for most models, asymptotic pivots or the bootstrap are the practical tools.
- The CI for the variance derived from the chi-squared pivot is asymmetric and not centred at s^2; do not report it as "$s^2 \pm$ margin".
- Pivotal CIs for transformations (e.g., log-odds) can be constructed on the transformed scale and back-transformed; this preserves exact coverage if the pivot is exact.
- Not every test statistic is a pivot – the Wald statistic is not, in general, exactly pivotal, which is why its CIs are approximate.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision

- Import, joins, and missingness with dplyr